

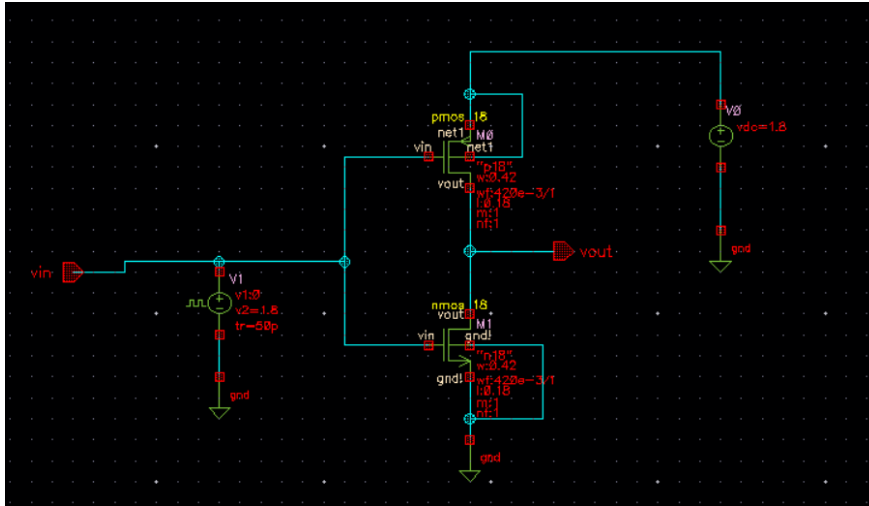
DIGITAL VLSI DESIGN

LAB REPORT

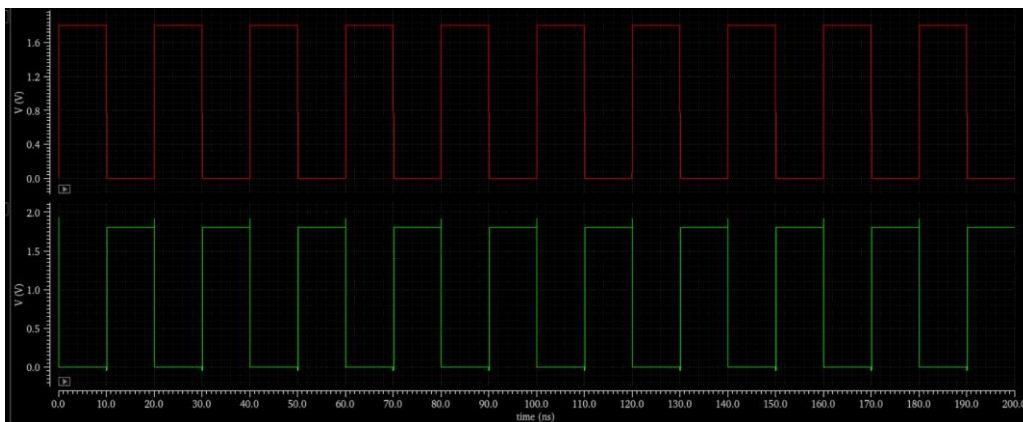
NAME:S.DHYAN
SRN- PES2UG24EC095
SEM- 4
SEC-"B"

1.CMOS INVERTER

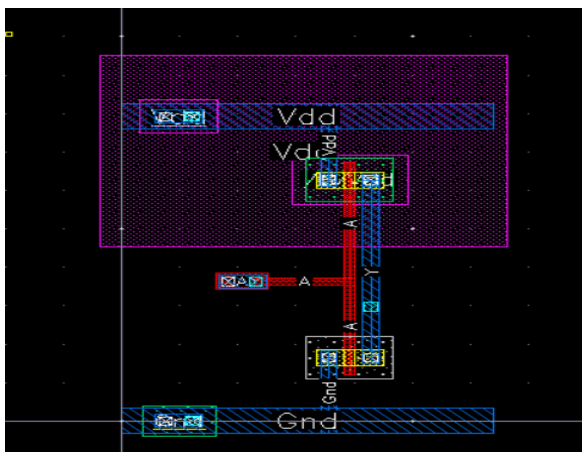
SCHEMATIC:



TRANSIENT ANALYSIS:

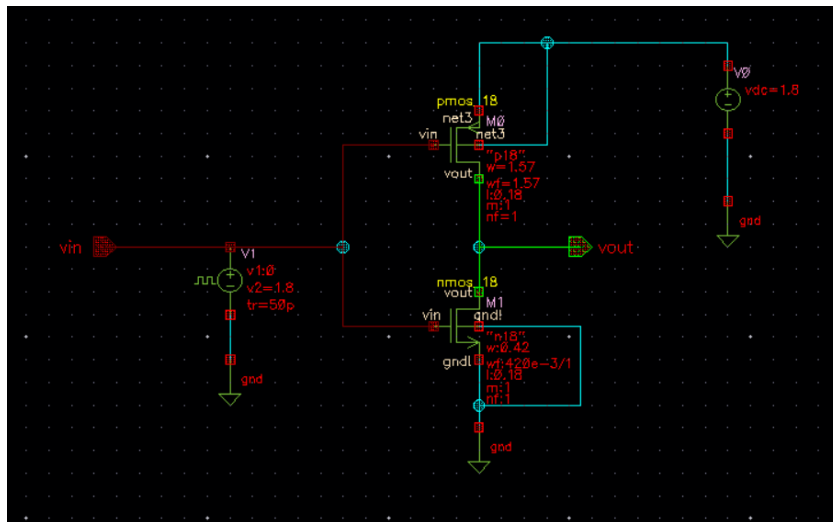


LAYOUT:



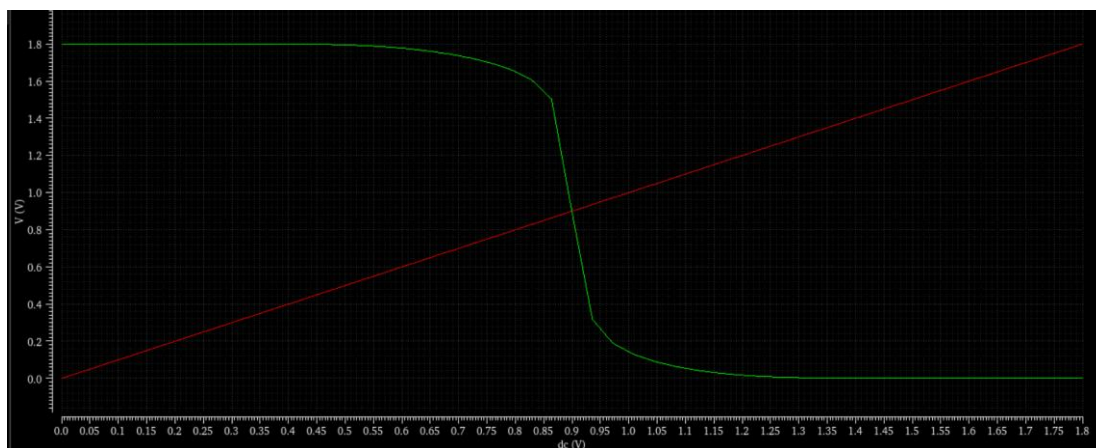
2.SYMMETRIC CMOS INVERTER:

Schematic:

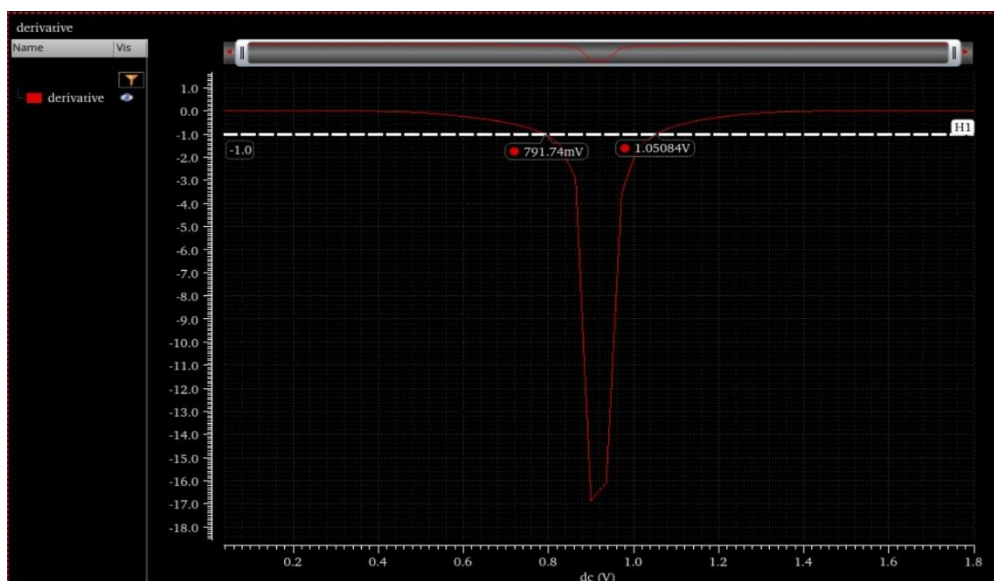


$(w/l)_n = (0.42/0.18)$
 $(w/l)_p = (1.57/0.18)$
 $NML = v_{il} - v_{ol} = 791.7\text{mV}$
 $NMH = v_{oh} - v_{ih} = 750\text{mV}$

DC analysis:

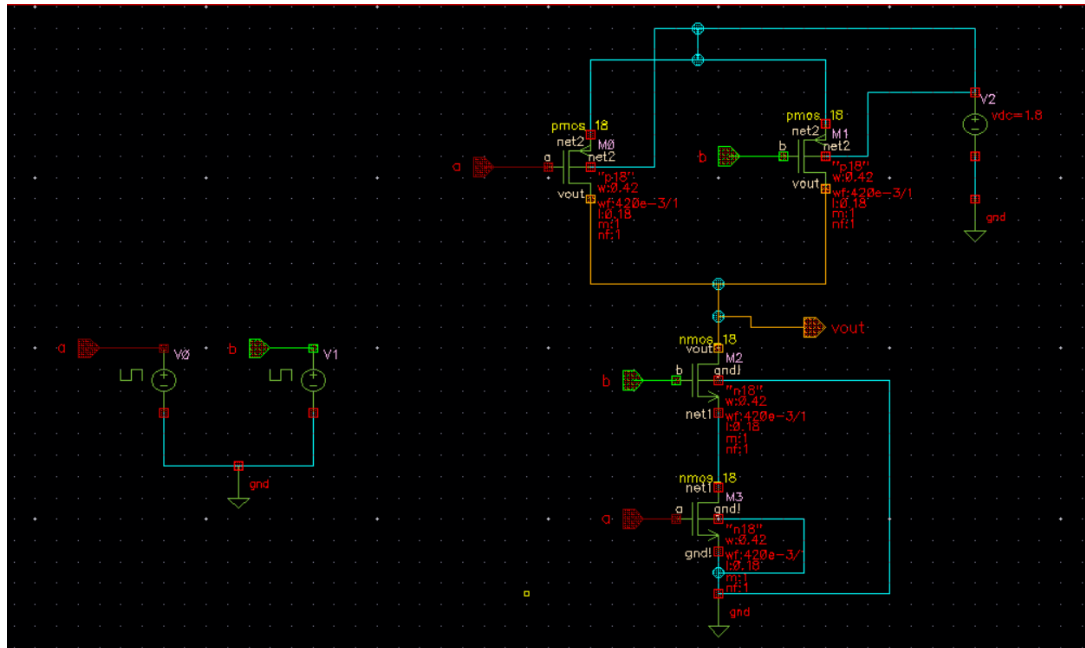


Noise margin analysis:

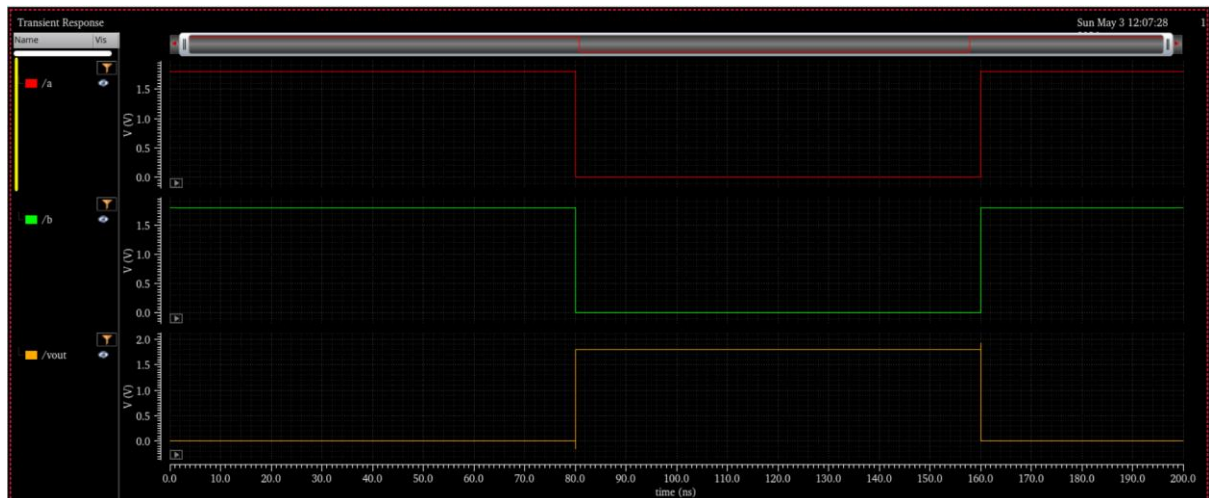


3.NAND GATE :

Schematic:



Transient analysis:



DELAYS:

1. $T_{plh_a} = 20.0208$ ns

2. $t_{plh_b} = 25.1098$ ns

3. $T_{plh_A} = 20.0251$ ns

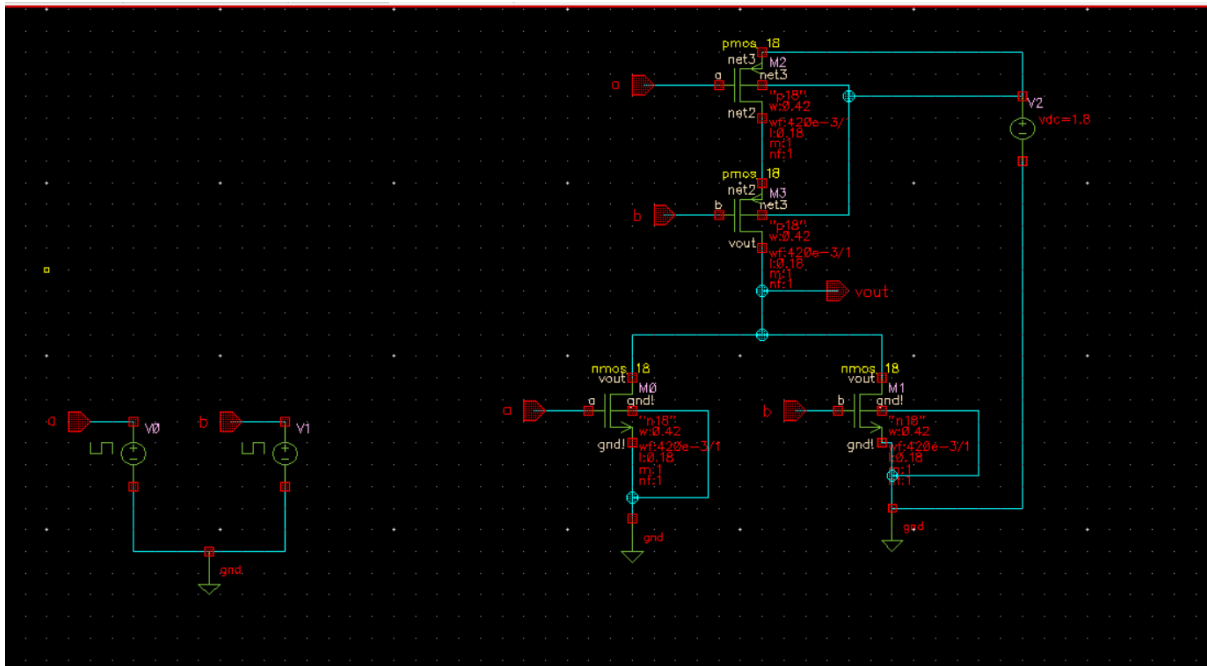
4. $T_{plh_B} = 10.0208$ ns

5. $T_{rise} = 21.6$ ps

6. $T_{fall} = 25.5$ ps

4.NOR GATE :

Schematic:



Transient analysis:

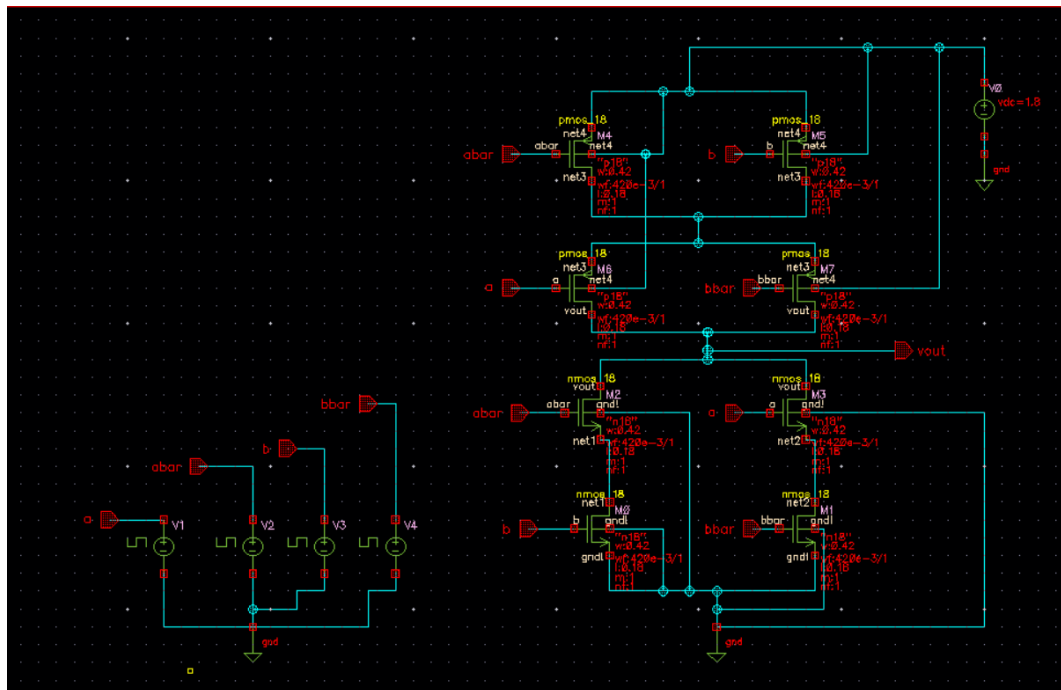


DELAYS:

1. $T_{phl_A} = 9.98351 \text{ ns}$
2. $T_{phl_B} = 16.4872 \text{ ps}$
3. $T_{rise} = 79.2 \text{ ns}$
4. $T_{plh_A} = 72.4638 \text{ ps}$
5. $T_{plh_B} = 20.0725 \text{ ns}$
6. $T_{fall} = 16.9 \text{ ns}$

5.XNOR GATE:

Schematic:



Transient analysis:

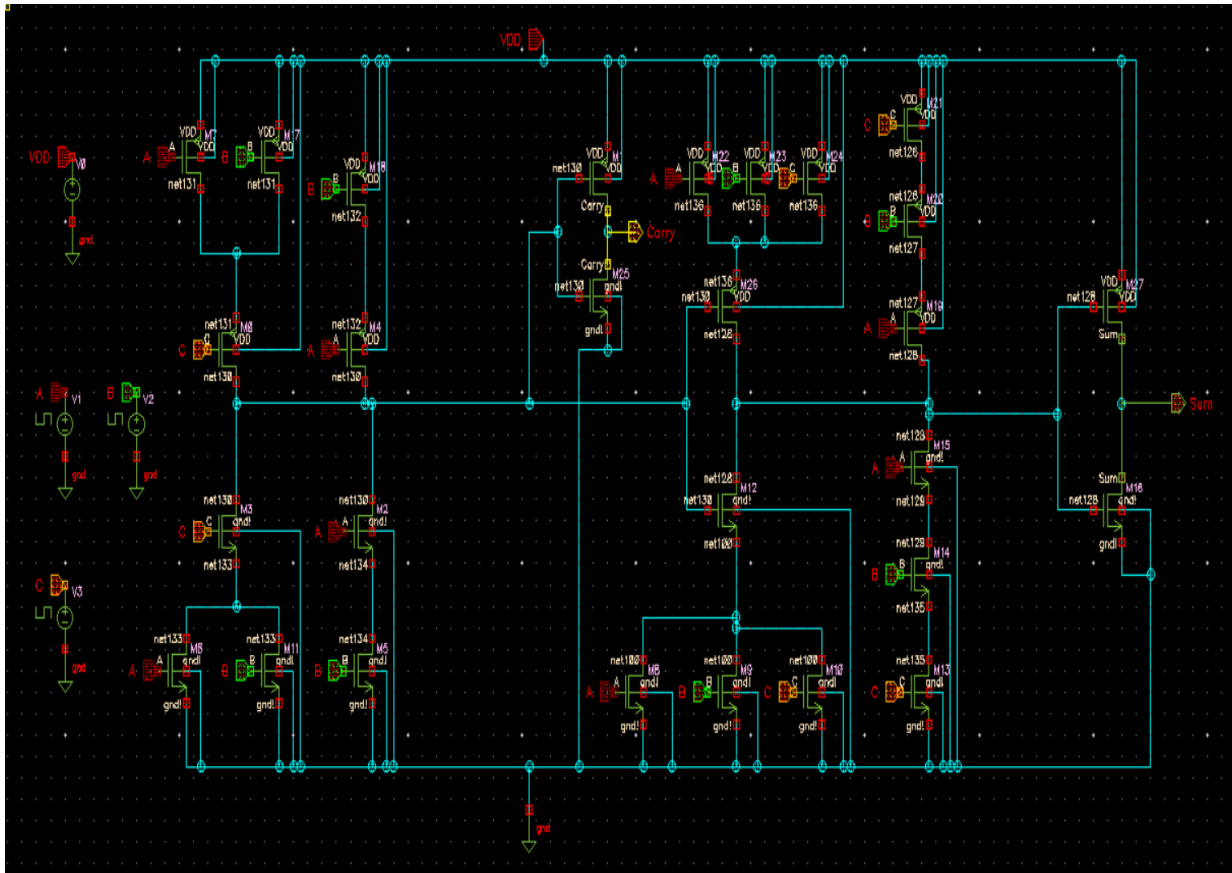


Delays:

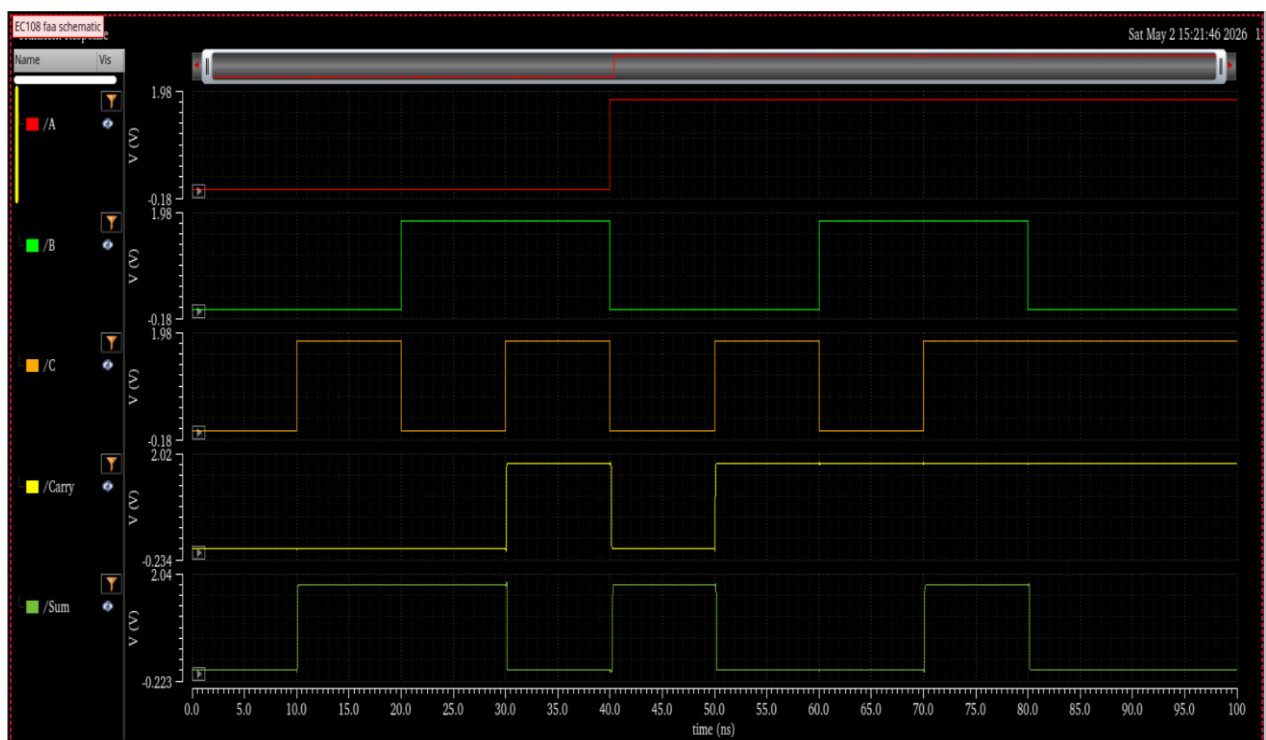
1. $T_{phl_A} = 9.96856 \text{ ns}$
2. $T_{phl_B} = 31.4366 \text{ ps}$
3. $T_{rise} = 147.8 \text{ ps}$
4. $T_{plh_A} = 9.86472 \text{ ns}$
5. $T_{plh_B} = 10.1353 \text{ ns}$
6. $T_{fall} = 29.5 \text{ ps}$

6.full adder using mirror logic:

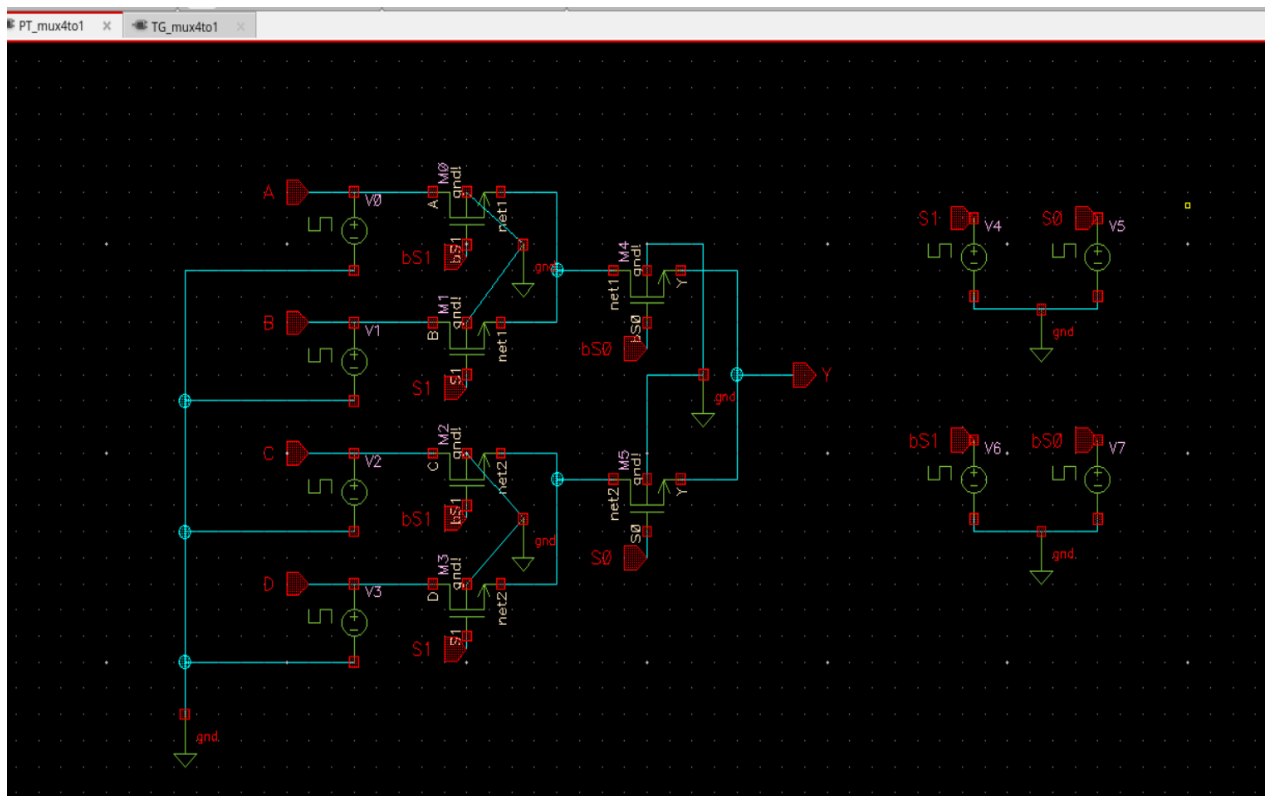
Schematic:



Transient:



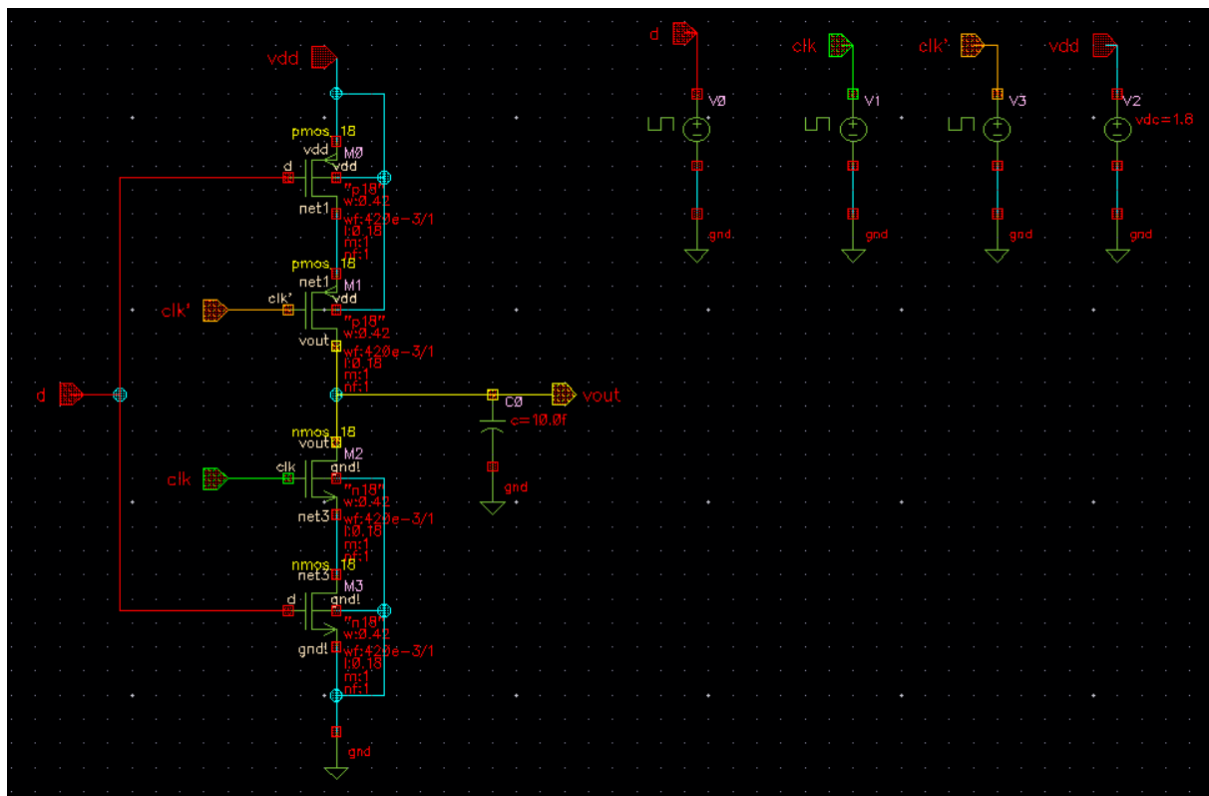
7. 4:1 MUX (Pass Transistor Logic):



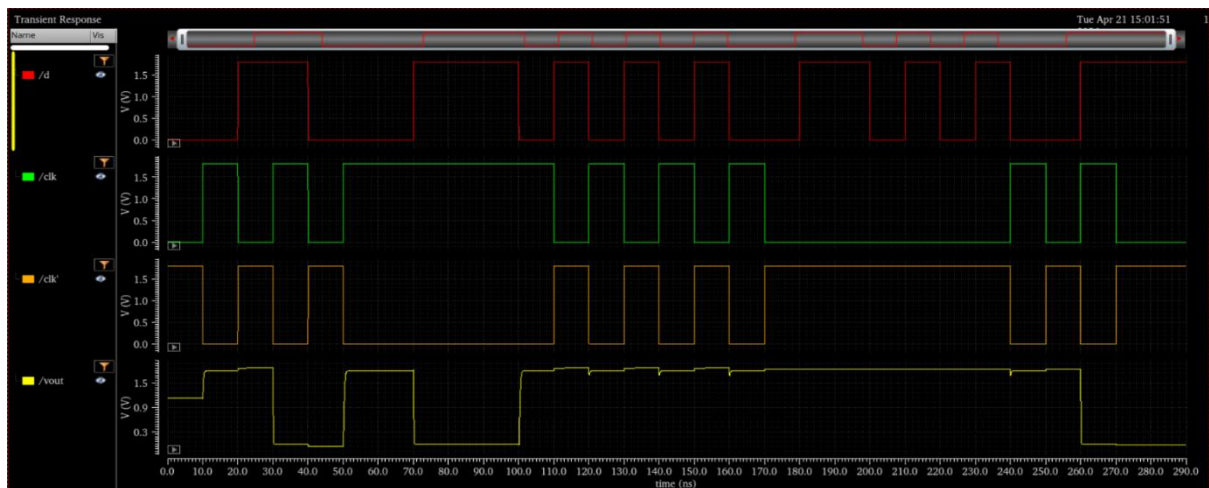
Transient analysis:



Schematic:

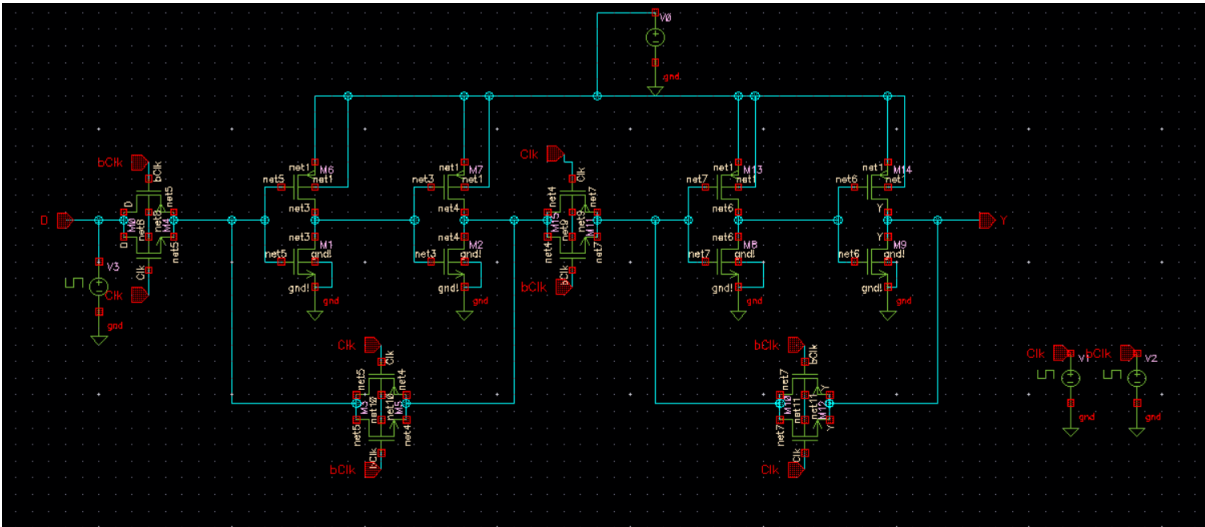


Transient analysis:

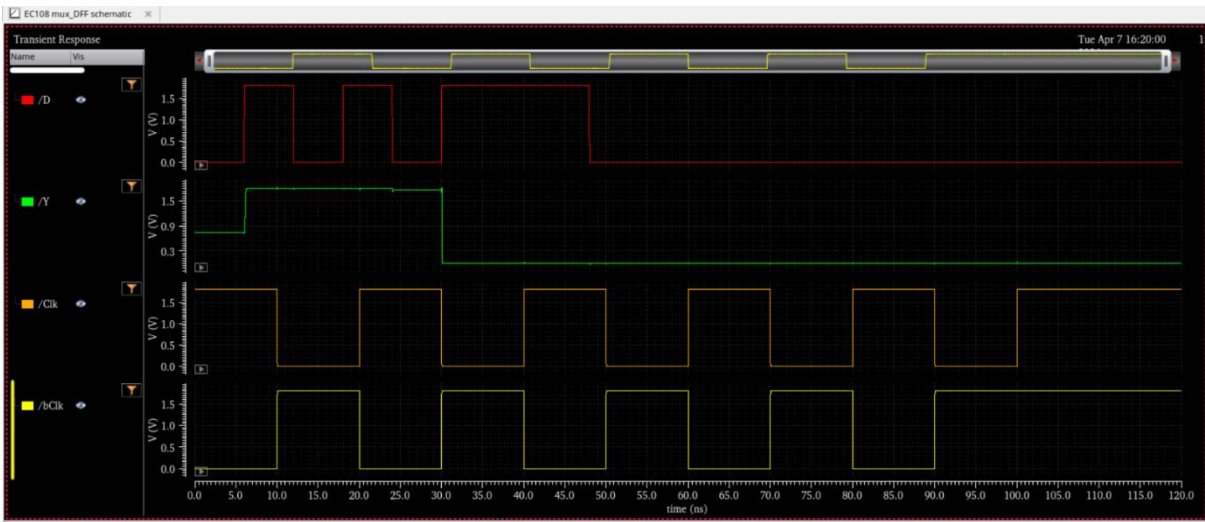


9. MUX based D Flip-Flop

Schematic:

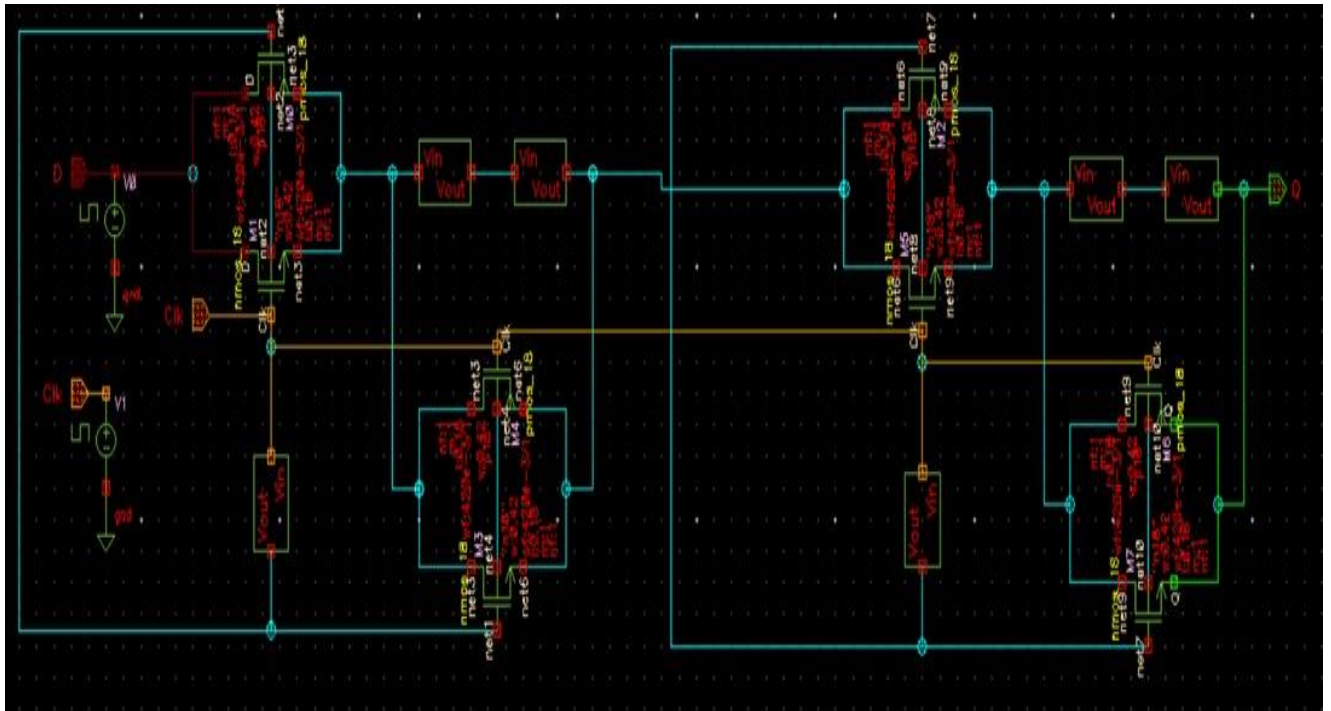


Transient analysis:

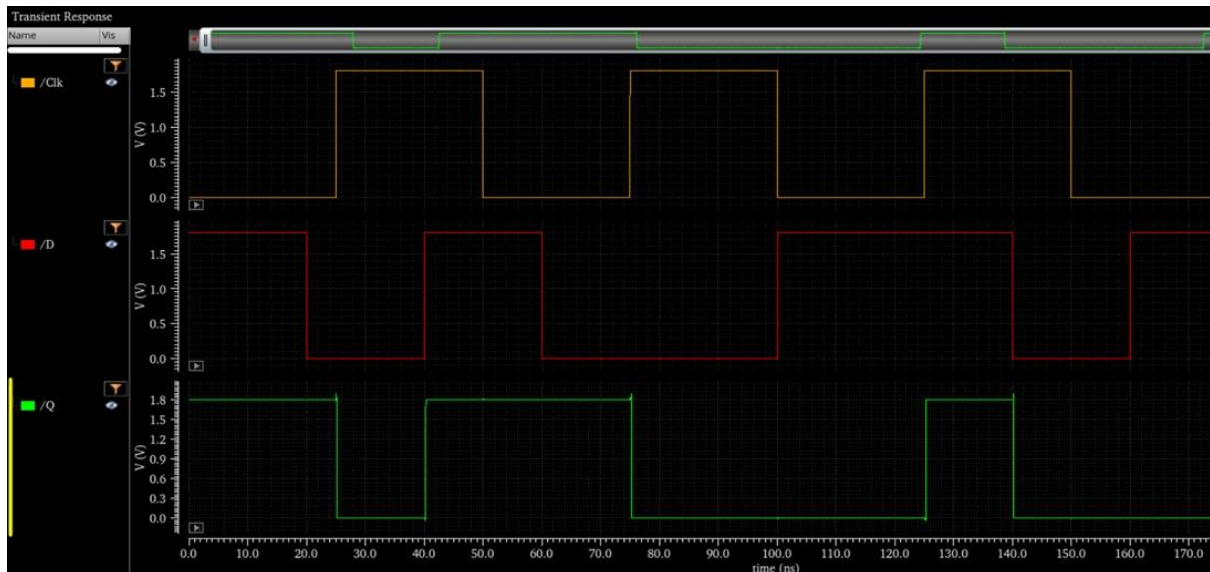


10. D flipflop using transmission gate:

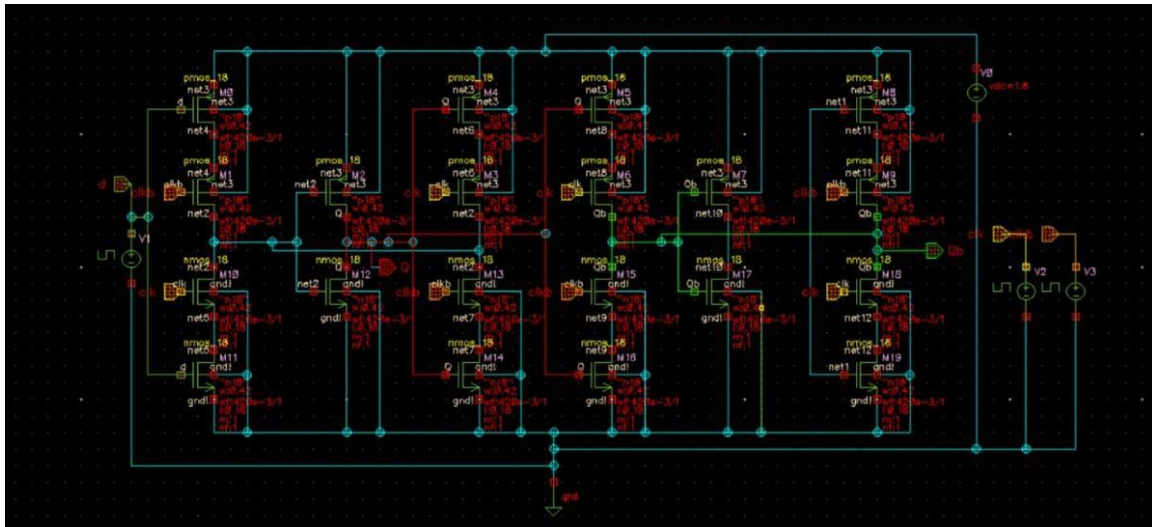
Schematic:



Transient analysis:



Schematic:

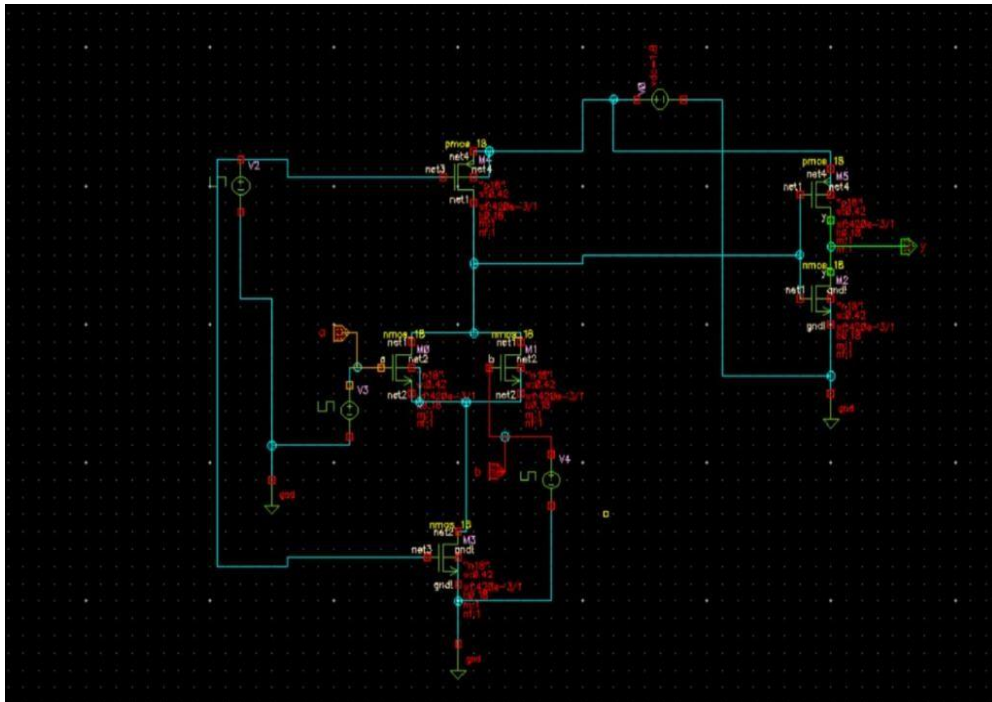


Transient analysis:

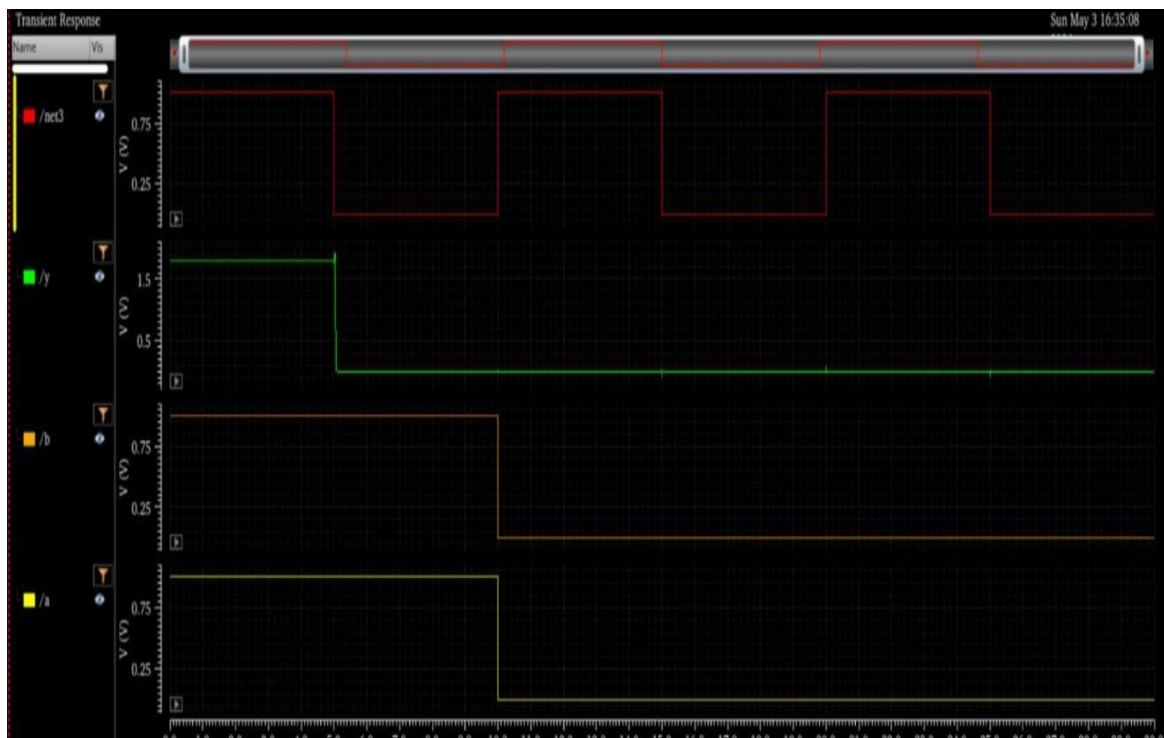


12.Domino logic

Schematic :

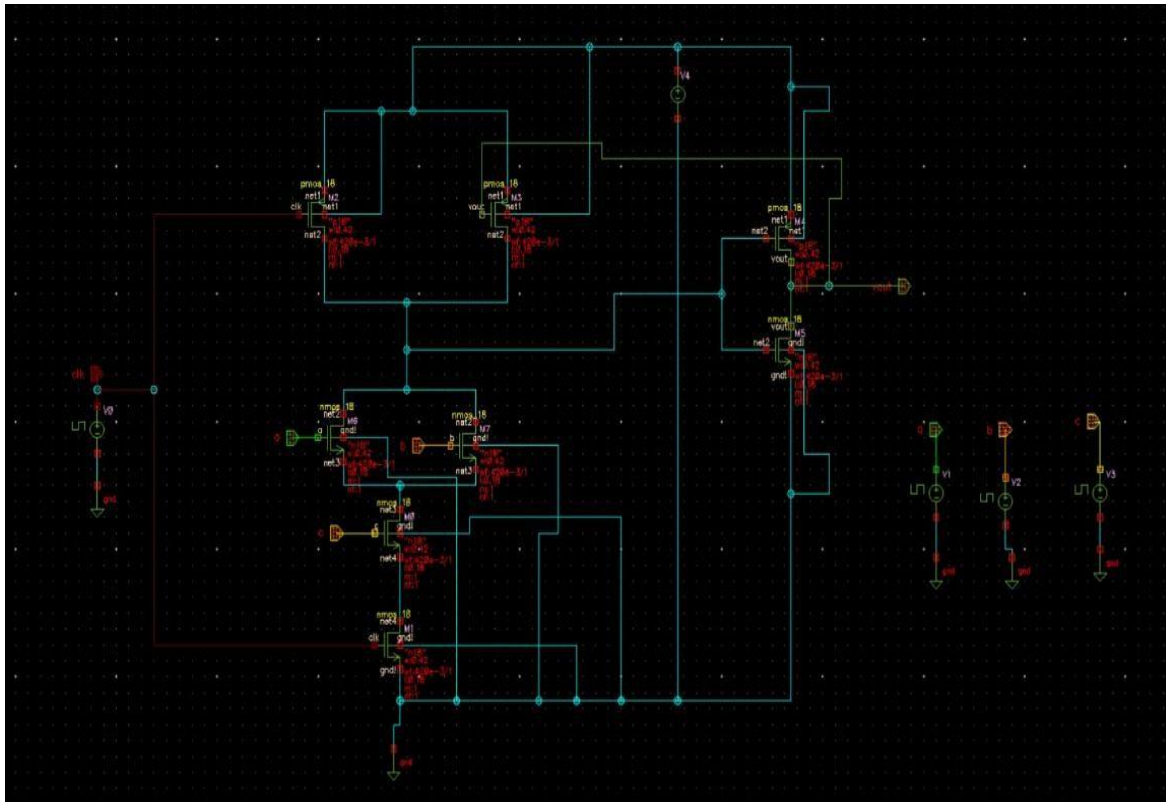


Transient analysis:



13.Domino logic using keeper circuit

Schematic :



Transient analysis:

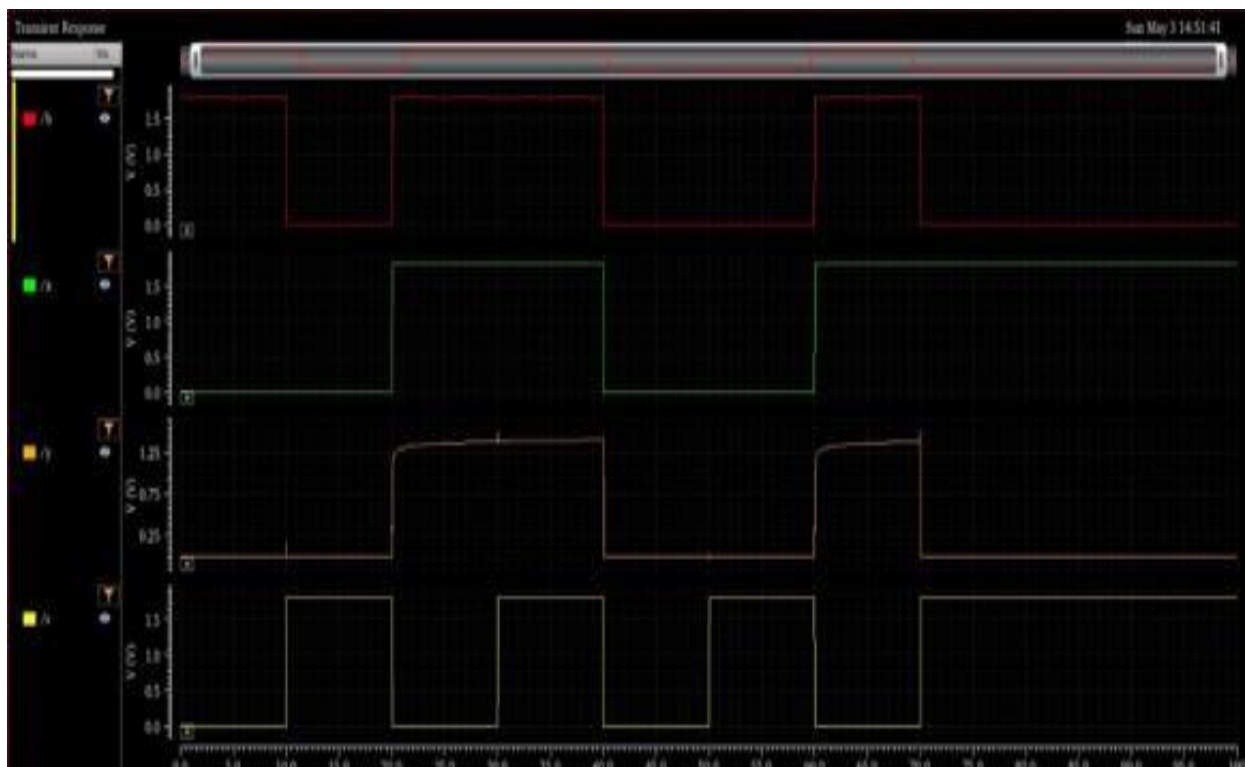


14.Pass transistor logic

Schematic

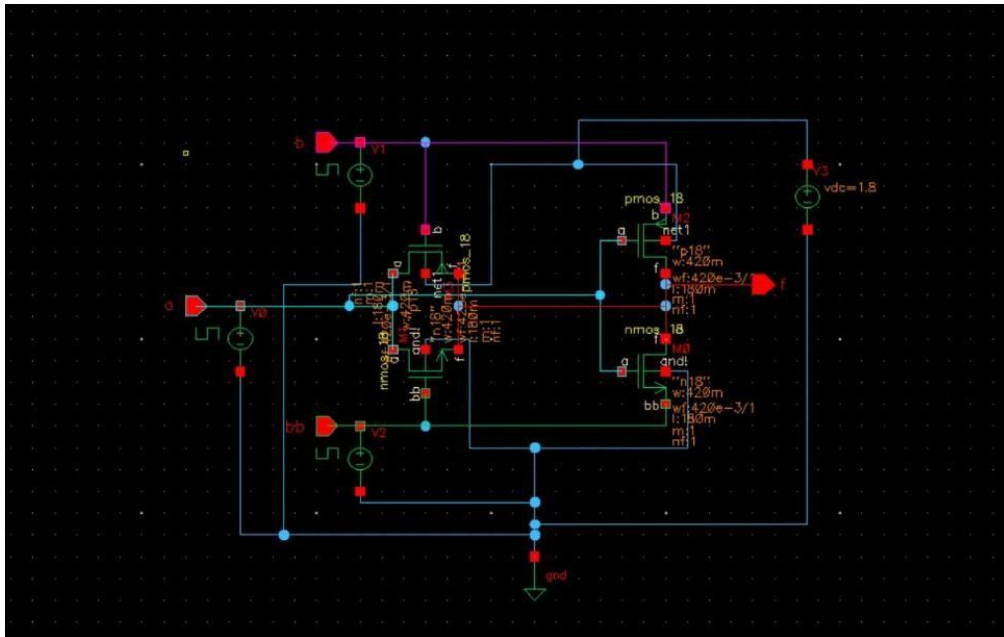


Transient analysis:



15. transmission gate logic

Schematic:



Transient analysis:

